



UE24MA241B LINEAR ALGEBRA AND ITS APPLICATIONS

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LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION

slides by Renna Sultana, Department of Science and Humanities



LINEAR ALGEBRA AND ITS APPLICATIONS

GAUSSIAN ELIMINATION:



1. Check for consistency and solve the following system of equations if consistent:

$$\begin{aligned} \text{(i)} \quad & x_1 + x_2 - 2x_3 + 3x_4 = 4 \\ & 2x_1 + 3x_2 + 3x_3 - x_4 = 3 \\ & 5x_1 + 7x_2 + 4x_3 + x_4 = 5 \end{aligned} \quad [A:b] = \begin{pmatrix} 1 & 1 & -2 & 3 & : & 4 \\ 2 & 3 & 3 & -1 & : & 3 \\ 5 & 7 & 4 & 1 & : & 5 \end{pmatrix} \begin{matrix} \downarrow R_2 - 2R_1 \\ \downarrow R_3 - 5R_1 \end{matrix} \rightarrow$$
$$\begin{bmatrix} 1 & 1 & -2 & 3 & : & 4 \\ 0 & 1 & 7 & -7 & : & -5 \\ 0 & 2 & 14 & -14 & : & -15 \end{bmatrix} \xrightarrow{\begin{matrix} \downarrow R_2 - R_1 \\ \downarrow R_3 - 2R_2 \end{matrix}} \begin{bmatrix} 1 & 1 & -2 & 3 & : & 4 \\ 0 & 1 & 7 & -7 & : & -5 \\ 0 & 0 & 0 & 0 & : & -5 \end{bmatrix}$$

This gives $0 = -5$ which is not possible.

Also $r(A)=2$ and $r[A:b]=3$

System is **inconsistent** and has **no solution**

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GAUSSIAN ELIMINATION:



$$(ii) \begin{aligned} x_1 + 2x_2 + x_3 &= 3 \\ 2x_1 + 5x_2 - x_3 &= -4 \\ 3x_1 - 2x_2 - x_3 &= 5 \end{aligned} \quad [A:b] = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 5 & -1 & -4 \\ 3 & -2 & -1 & 5 \end{pmatrix}$$

$$\downarrow \begin{matrix} R_2 - 2R_1 \\ R_3 - 3R_1 \end{matrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & -3 & -10 \\ 0 & -8 & -4 & -4 \end{bmatrix} \quad \downarrow \begin{matrix} R_3 + 8R_2 \end{matrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & -3 & -10 \\ 0 & 0 & -28 & -84 \end{bmatrix}$$

$$\Rightarrow \begin{cases} x_1 + 2x_2 + x_3 = 3 \\ x_2 - 3x_3 = -10 \\ -28x_3 = -84 \end{cases}$$

$r(A)=r[A:b]=3=n$. System is **consistent** and has **a unique solution**.

$$(x_1, x_2, x_3) = (2, -1, 3)$$

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GAUSSIAN ELIMINATION:



$$\begin{aligned} \text{(iii)} \quad & 2x - 3y + 2z = 1 \\ & 5x - 8y + 7z = 1 \\ & y - 4z = 3 \end{aligned}$$

$$\begin{pmatrix} 2 & -3 & 2:1 \\ 5 & -8 & 7:1 \\ 0 & 1 & -4:3 \end{pmatrix} \xrightarrow{\substack{R_2 \leftarrow \left(\frac{5}{2}\right)R_1 \\ R_3 \leftarrow \left(\frac{5}{2}\right)R_1}} \begin{pmatrix} 2 & -3 & 2:1 \\ 0 & -1/2 & 2:-3/2 \\ 0 & 1 & -4:3 \end{pmatrix} \xrightarrow{\substack{R_2 \leftarrow \left(\frac{5}{2}\right)R_1 \\ R_3 \leftarrow \left(\frac{5}{2}\right)R_1}} \begin{pmatrix} 2 & -3 & 2:1 \\ 0 & -1/2 & 2:-3/2 \\ 0 & 0 & 0:0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} 2x - 3y + 2z = 1 \\ (1/2)y + 2z = -3/2 \end{cases}$$

$r(A)=r(A:b)=2 < n(=3)$ hence system is **consistent**
and has **infinite number of solutions**.
i.e **$(x, y, z) = (5k+5, 4k+3, k)$**

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GAUSSIAN ELIMINATION:



2. Find all values of a for which the resulting linear system has (a) no solution (b) a unique solution and (c) infinitely many solutions:

$$x + y - z = 2$$

$$x + 2y + z = 3$$

$$x + y + (a^2 - 5)z = a$$

$$[A:b] = \begin{pmatrix} 1 & 1 & -1 & : & 2 \\ 1 & 2 & 1 & : & -3 \\ 1 & 1 & a^2 - 5 & : & a \end{pmatrix} \xrightarrow[R_3 - R_1]{R_2 - R_1} \begin{pmatrix} 1 & 1 & -1 & : & 2 \\ 0 & 1 & 2 & : & -5 \\ 0 & 0 & a^2 - 4 & : & a - 2 \end{pmatrix}$$

- (a) System has **no solution** if $a = -2$ (when $r(A) \neq r(A:b)$)
(b) System has **a unique solution** if $a \neq \pm 2$ (when $r(A) = r(A:b) = 3 = n$)
(c) System has **infinitely many solutions** if $a = 2$ (when $r(A) = r(A:b) = 2 < n$)

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GAUSSIAN ELIMINATION:



3. Find an equation relating a , b and c so that the linear system $x + 2y - 3z = a$ is consistent for any values of a , b and c that satisfy that equation. When $(a,b,c)=(2,3,9)$, then what is the solution of the system.
- $$\begin{aligned} 2x + 3y + 3z &= b \\ 5x + 9y - 6z &= c \end{aligned}$$

$$\begin{pmatrix} 1 & 2 & -3 & : & a \\ 2 & 3 & 3 & : & b \\ 5 & 9 & -6 & : & c \end{pmatrix} \xrightarrow[R_3-5R_1]{R_2-2R_1} \begin{pmatrix} 1 & 2 & -3 & : & a \\ 0 & -1 & 9 & : & b-2a \\ 0 & -1 & 9 & : & c-5a \end{pmatrix}$$

$$\xrightarrow{R_3-R_2} \begin{pmatrix} 1 & 2 & -3 & : & a \\ 0 & -1 & 9 & : & b-2a \\ 0 & 0 & 0 & : & c-b-3a \end{pmatrix}$$

The given linear system will be **consistent** if a, b, c satisfy the relation $c-b-3a=0$.

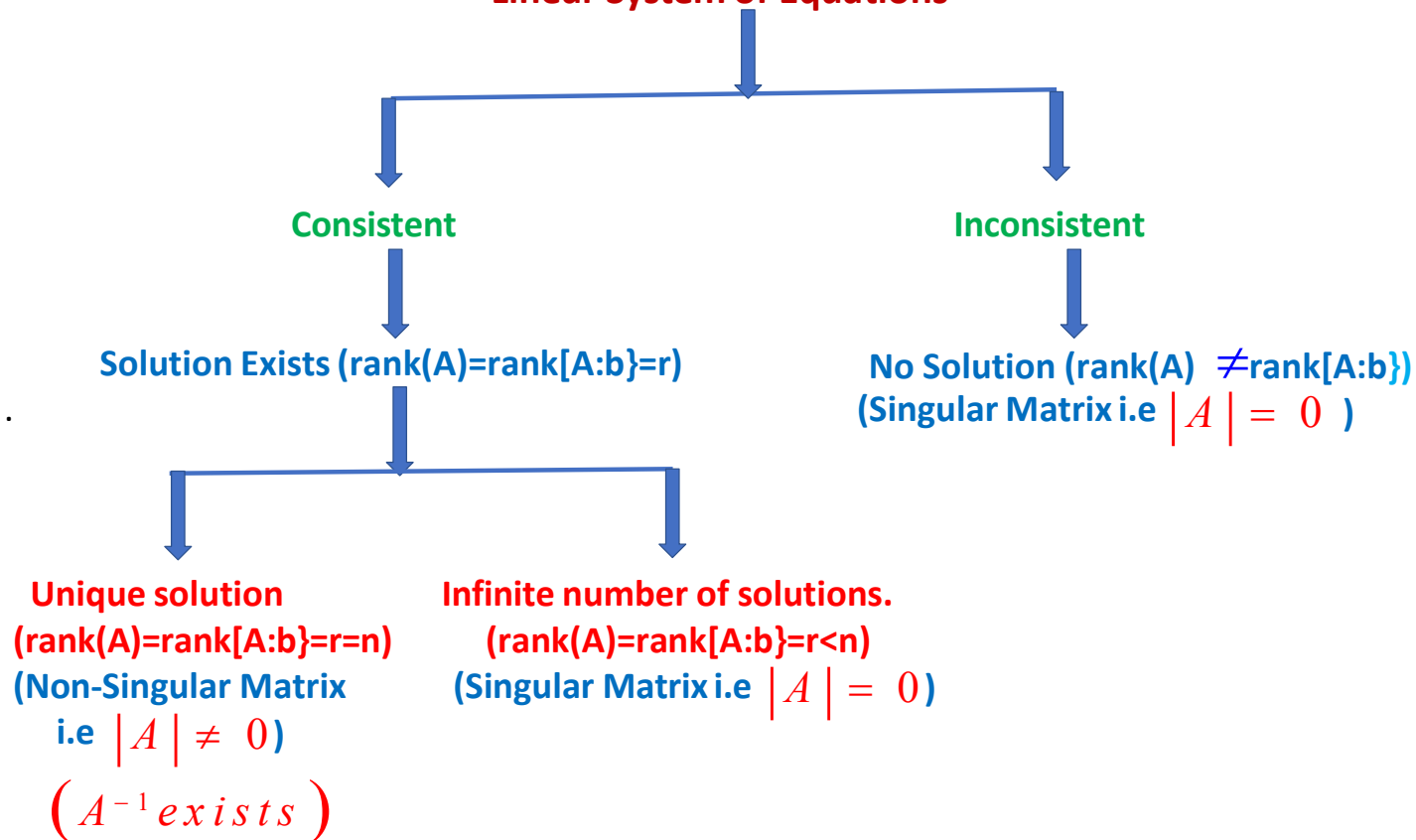
When $(a, b, c)=(2, 3, 9)$, then the solution of the system is $(x, y, z)=(-15k, 9k+1, k)$

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GAUSSIAN ELIMINATION:



Linear System of Equations





THANK YOU

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